Group - 7

The ‘Drim’ Programming Language

Bringing the Classroom Joke to Life via Language Implementation

Course: SWE 4304 (Software Project Lab - I)

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What Are We Making?

The Drim Concept

Drim is a **custom-built**, **interpreted** programming language designed to operate in a non-GUI (command-line) environment.

The CLI Interpreter

A **command-line interpreter** that reads .drim source files and executes them step by step.

Recap on The Architecture

Lexer (The Scanner)

- Reads raw text input and converts it into "Tokens"
  - transforming "drimming" into a recognizable loop command.

Parser (The Architect)

- Organizes the generated tokens into an list of Abstract Syntax Tree (AST)s to understand logic and math hierarchy.

Interpreter (The Engine)

- Traverses the Abstract Syntax Tree, executing the code in real-time. This is where Drim comes to life!

Features

I/O

- Input → “drim()”
- Output → “wake()” / “wakef()”

Math Operations

Supports most arithmetic operations including basic (+, -, *, /, %) and Bitwise (and, or, not, >> , <<)

Data Types

Supports INT, FLOAT, STRING, BOOLEAN as data types

Conversion Functions

No need to memorize how to convert between different unit!

Dynamic Typing

Variables seamlessly switch types (e.g., Int ↔ String) on the fly, offering flexibility in data handling.

Branching (If- Else)

Supports branching of block executions using “if-else” statements.

Physics Formula Calculation

Forgot how to calculate data like final velocity? We’ve got that covered! :)

Features

Loops

Loops for multiple iteration of blocks. (drimming)

Scoped Variable

Variables lives in their own block.

Custom User Defined Functions

Supports custom function calling with parameters and dynamic return values.

String Interpolation

Formatted output, like

- wake(“Hello {name}”)

Data Structure Library

Custom built built-in data types like

- Arrays
- Stack
- Queue

Contributions

Akhiar

- Basic Architecture & I/O
- Variable Management & Automatic Type Casting
- Multi-assignment syntax
- Branching and Loops
- Arrays + Math Modulo (%)

Emon

- Mathematical Operations
- Scoped Variable
- User-defined Functions
- Physics Function Engine
- String Interpolation + Output Formatting + Bool Data-Type

Tamim

- Data Structures Library
  - Stacks
  - Queues
- Unit Conversion Functions
- Internal Calculation Precision for Int & Float

Technologies Used

Programming Language

C++

Build System

CMake
Cross-platform Make

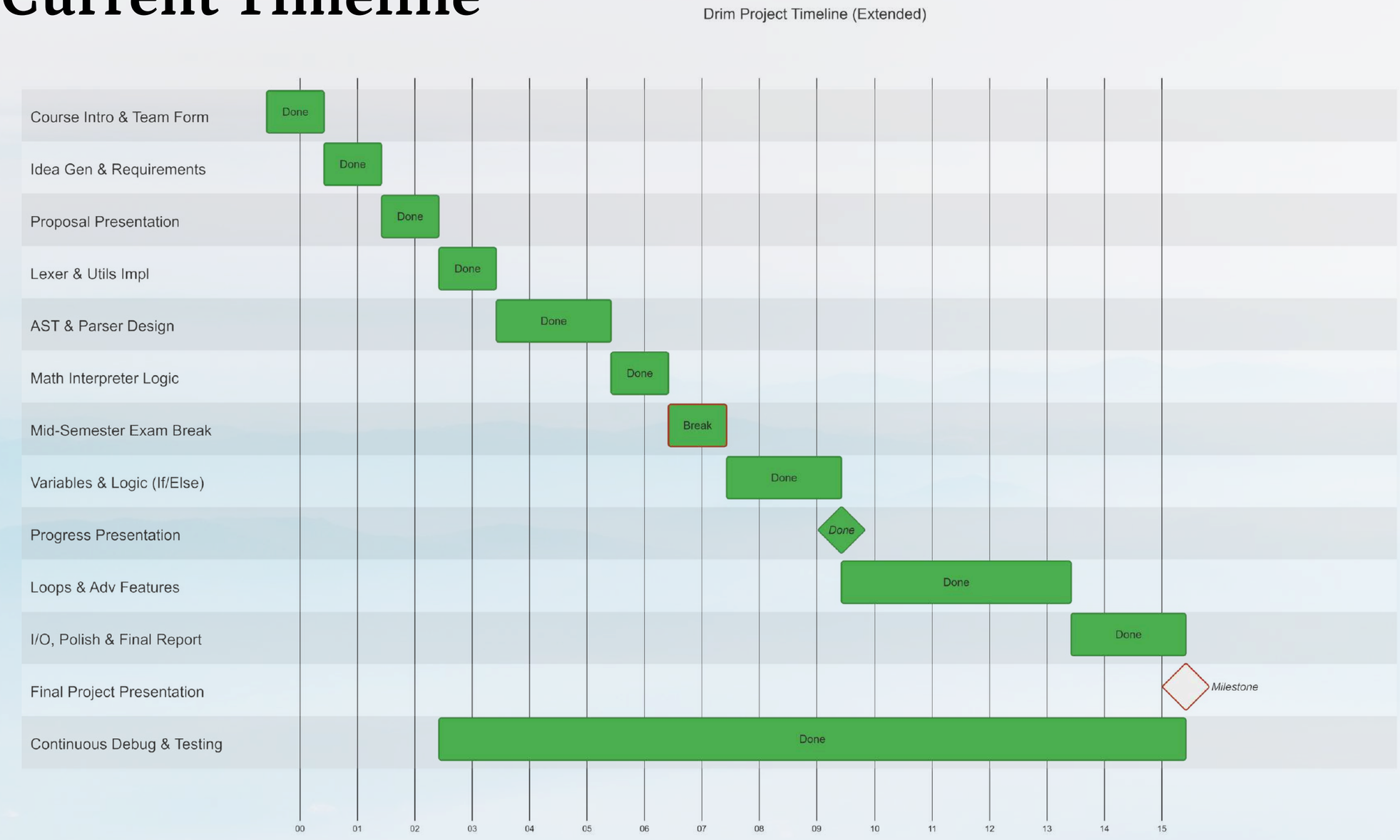
Version Control

IDE

Visual Studio Code

Diagramming

Current Timeline



Demo

QNA